

C1002 — Programming Fundamentals | PF Lab Tasks

FAST — National University of Computer and Emerging Sciences Karachi campus

Department of Software Engineering

PF LAB TASKS

Course Code: C1002

Subject: Programming Fundamentals

Section: C

Instructor: Mr. Shayan Ali

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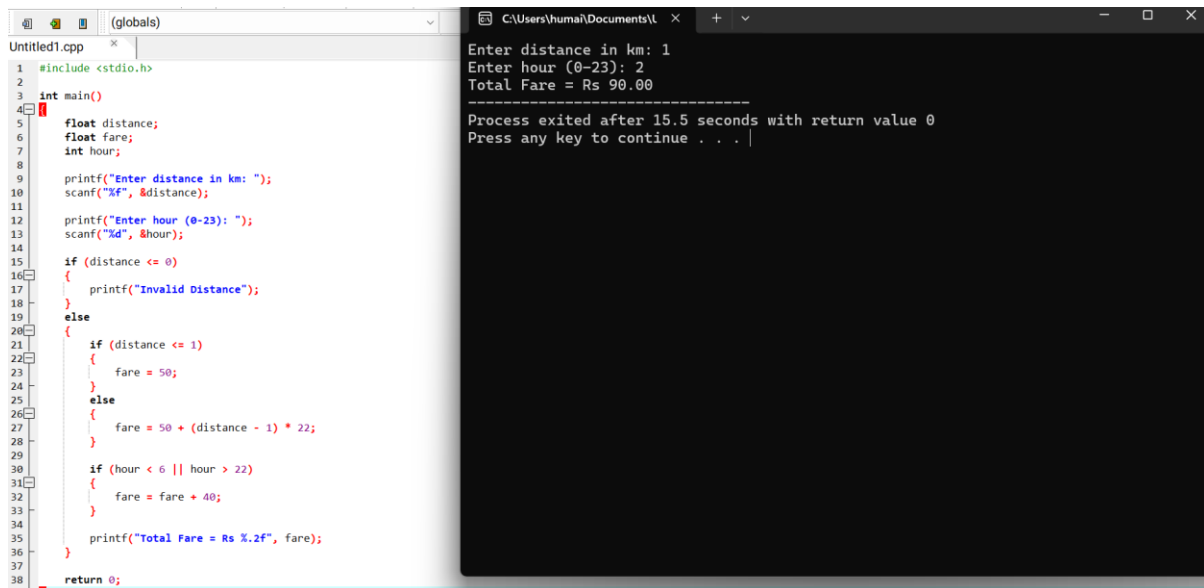
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Task 1 — The Rickshaw Fare Calculator

Write a C program that takes the distance traveled in kilometers as a float and the hour of the day as an integer from 0–23 using scanf. The base fare is Rs 50 for the first 1 km, and every kilometer after the first costs Rs 22. If the hour is before 6 or after 22, add a Rs 40 night surcharge. If the distance is 0 or negative, display "Invalid Distance" and stop the program. Finally, display the total fare in Rupees up to 2 decimal places. Remember that the first 1 km already costs Rs 50, so the Rs 22 per-kilometer charge applies only to the distance beyond 1 km

The image shows a code editor window on the left and a terminal window on the right. The code editor contains a C program named 'Untitled1.cpp' that calculates a total fare based on distance and hour. The program includes `<stdio.h>`, declares `float distance;`, `float fare;`, and `int hour;` in the `main()` function. It prompts the user to enter distance and hour, then uses `scanf` to read the inputs. A series of `if-else` statements determine the fare: if distance is less than or equal to 0, it's invalid; if distance is between 1 and 5, fare is 50; otherwise, fare is 50 plus 22 times the distance minus 1. Additionally, if the hour is between 6 and 22, the fare is increased by 40. Finally, it prints the total fare using `printf` and returns 0. The terminal window shows the execution of this program with inputs 1 for distance and 2 for hour, resulting in a total fare of Rs 90.00. It also shows the process exit message and a prompt to press any key to continue.

Task 2 — The Grocery Store Discount Slip

Write a C program that takes the total bill amount as a float and membership status as an integer, where 1 means member and 0 means non-member. If the bill is below Rs 500, no discount is given. For bills from Rs 500 to Rs 1999, members receive a 10% discount while non-members receive 5%. For bills of Rs 2000 or more, members receive 15% while non-members receive 8%. Display both the discount amount and the final payable amount up to 2 decimal places. The discount must be determined using both the bill range and membership status together

```
1 #include <stdio.h>
2
3 int main()
4 {
5     float bill, discount, payable;
6     int member;
7
8     printf("Enter total bill: ");
9     scanf("%f", &bill);
10
11     printf("Enter membership status (1=Member, 0=Non-member): ");
12     scanf("%d", &member);
13
14     discount = 0;
15
16     if (bill < 500)
17     {
18         discount = 0;
19     }
20     else if (bill < 2000)
21     {
22         if (member == 1)
23         {
24             discount = bill * 0.10;
25         }
26         else
27         {
28             discount = bill * 0.05;
29         }
30     }
31     else
32     {
33         if (member == 1)
34         {
35             discount = bill * 0.15;
36         }
37         else
38         {
39             discount = bill * 0.08;
40         }
41     }
42
43     payable = bill - discount;
44
45     printf("Discount = Rs %.2f\n", discount);
46     printf("Final Payable Amount = Rs %.2f", payable);
47
48     return 0;
49 }
```

Output:

```
Enter total bill: 2000
Enter membership status (1=Member, 0=Non-member): 1
Discount = Rs 300.00
Final Payable Amount = Rs 1700.00
-----
Process exited after 16.02 seconds with return value 0
Press any key to continue . . .
```

Task 3 — The Exam Grade Slip

Write a C program that takes marks obtained as an integer out of 100 using scanf. If the marks are below 0 or above 100, display "Invalid Marks" and stop. Assign grades according to these ranges: 90–100 = A+, 80–89 = A, 70–79 = B, 60–69 = C, 50–59 = D, and below 50 = F. If the grade is F, display "Result: Fail"; otherwise display "Result: Pass". Finally, print both the grade and result message. Check the grade boundaries from highest to lowest.

The image shows a C++ IDE with two windows. The top window, titled 'Untitled1.cpp', contains the following code:

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int marks;
6
7     printf("Enter marks: ");
8     scanf("%d", &marks);
9
10    if (marks < 0 || marks > 100)
11    {
12        printf("Invalid Marks");
13    }
14    else
15    {
16        if (marks >= 90)
17        {
18            printf("Grade: A\n");
19            printf("Result: Pass");
20        }
21        else if (marks >= 80)
22        {
23            printf("Grade: A\n");
24            printf("Result: Pass");
25        }
26        else if (marks >= 70)
27        {
28            printf("Grade: B\n");
29            printf("Result: Pass");
30        }
31        else if (marks >= 60)
32        {
33            printf("Grade: C\n");
34            printf("Result: Pass");
35        }
36        else if (marks >= 50)
37        {
38            printf("Grade: D\n");
39            printf("Result: Pass");
40        }
41    }
42 }
43
44 return 0;
```

The bottom window shows the execution output:

```
Enter marks: 70
Grade: B
Result: Pass
-----
Process exited after 14.95 seconds with return value 0
Press any key to continue . . .
```

Task 4 — The Water Tank Timer

Write a C program that takes the tank capacity in liters as an integer, the current water level in liters as an integer, and the motor fill rate in liters per minute as a float. Calculate how many liters are still required to fill the tank. If the current water level is greater than or equal to the capacity, display "Tank Already Full" and stop. Otherwise, calculate the required time by dividing the remaining liters by the fill rate. Electricity costs Rs 3.50 for every full or partial minute, so the required time must be rounded upward to the next whole minute using arithmetic and casting only, without library rounding functions. Finally, display the required time to 2 decimal places and the billed electricity cost.

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int capacity, current, minutes;
6     float rate, remaining, time, cost;
7
8     printf("Enter tank capacity in liters: ");
9     scanf("%d", &capacity);
10
11    printf("Enter current water level in liters: ");
12    scanf("%d", &current);
13
14    printf("Enter motor fill rate (liters/minute): ");
15    scanf("%f", &rate);
16
17    if (current >= capacity)
18    {
19        printf("Tank Already Full");
20    }
21    else
22    {
23        remaining = capacity - current;
24        time = remaining / rate;
25        minutes = (int)time;
26
27        if (time > minutes)
28        {
29            minutes++;
30        }
31
32        cost = minutes * 3.50;
33
34        printf("Required Time = %d minutes\n", minutes);
35        printf("Electricity Cost = Rs %.2f", cost);
36    }
37
38    return 0;
39 }
```

```
Enter tank capacity in liters: 100
Enter current water level in liters: 80
Enter motor fill rate (liters/minute): 20
Required Time = 1 minutes
Electricity Cost = Rs 3.50
-----
Process exited after 74.6 seconds with return value 0
Press any key to continue . . . |
```

Task 5 — The Mobile Load Card Vendor

Write a C program that takes the mobile load amount as a float, network code as an integer where 1 = Jazz, 2 = Telenor, and 3 = Ufone, and weekend status where 1 = weekend and 0 = weekday. If the amount is below Rs 100, there is no bonus. If the amount is between Rs 100 and Rs 499 and it is a weekend, the bonus is 10%, except for Ufone, which receives 5%. If the amount is between Rs 100 and Rs 499 on a weekday, the bonus is 5%. If the amount is Rs 500 or more and either the network is Jazz or it is a weekend, the bonus is 20%; otherwise, the bonus is 12%. Implement the rules using only nested if-else and follow the exact order. Finally, display the bonus and final loaded balance.

```

1 #include <stdio.h>
2
3 int main()
4 {
5     float amount, bonus, finalBalance;
6     int network, weekend;
7
8     printf("Enter load amount: ");
9     scanf("%f", &amount);
10
11     printf("Enter network (1=Jazz, 2=Telenor, 3=Ufone): ");
12     scanf("%d", &network);
13
14     printf("Enter weekend status (1=Weekend, 0=Weekday): ");
15     scanf("%d", &weekend);
16
17     bonus = 0;
18
19     if (amount < 100)
20     {
21         bonus = 0;
22     }
23     else
24     {
25         if (amount < 500)
26         {
27             if (weekend == 1)
28             {
29                 if (network == 3)
30                 {
31                     bonus = amount * 0.05;
32                 }
33                 else
34                 {
35                     bonus = amount * 0.10;
36                 }
37             }
38             else
39             {
40                 if (network == 3)
41                 {
42                     bonus = amount * 0.05;
43                 }
44                 else
45                 {
46                     bonus = amount * 0.10;
47                 }
48             }
49         }
50         else
51         {
52             if (network == 1 || weekend == 1)
53             {
54                 bonus = amount * 0.20;
55             }
56             else
57             {
58                 bonus = amount * 0.12;
59             }
60         }
61     }
62
63     finalBalance = amount + bonus;
64     printf("Bonus = Rs %.2f\n", bonus);
65     printf("Final Loaded Balance = Rs %.2f", finalBalance);
66
67     return 0;
68 }

```

```

Enter load amount: 300
Enter network (1=Jazz, 2=Telenor, 3=Ufone): 3
Enter weekend status (1=Weekend, 0=Weekday): 1
Bonus = Rs 15.00
Final Loaded Balance = Rs 315.00
-----
Process exited after 39.94 seconds with return value 0
Press any key to continue . . .

```

Task 6 — The Library Fine Calculator

Write a C program that takes the number of overdue days, book type, and priority membership status as integers. Book type 1 is Regular, 2 is Reference, and 3 is Rare, while priority membership is 1 for yes and 0 for no. Regular books cost Rs 5 per day for the first 7 overdue days and Rs 10 per day after that. Reference books cost Rs 15 per overdue day with no free days. Rare books cost Rs 30 per day and students overdue by more than 10 days must also receive the message "Banned from Borrowing". Priority members receive a 20% discount on the fine, except for Rare books where no discount is given. Display the fine up to 2 decimal places and display the ban message where applicable.

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int days, bookType, priority;
6     float fine;
7
8     printf("Enter overdue days: ");
9     scanf("%d", &days);
10
11     printf("Enter book type (1=Regular, 2=Reference, 3=Rare): ");
12     scanf("%d", &bookType);
13
14     printf("Enter priority membership (1=Yes, 0=No): ");
15     scanf("%d", &priority);
16
17     if (bookType == 1)
18     {
19         if (days <= 7)
20         {
21             fine = days * 5;
22         }
23         else
24         {
25             fine = 7 * 5 + (days - 7) * 10;
26         }
27
28         if (priority == 1)
29         {
30             fine = fine * 0.80;
31         }
32     }
33     else if (bookType == 2)
34     {
35         fine = days * 15;
36
37         if (priority == 1)
38         {
39             fine = fine * 0.80;
40         }
41     }
42     else if (bookType == 3)
43     {
44         fine = days * 30;
45     }
46     else
47     {
48         printf("Invalid Book Type");
49         return 0;
50     }
51
52     printf("Fine = Rs %.2f\n", fine);
53
54     if (bookType == 3 && days > 10)
55     {
56         printf("Banned from Borrowing");
57     }
58
59     return 0;
60 }
```

Untitled1.cpp

```
24 {
25     fine = 7 * 5 + (days - 7) * 10;
26 }
27
28 if (priority == 1)
29 {
30     fine = fine * 0.80;
31 }
32
33 }
34 else if (bookType == 2)
35 {
36     fine = days * 15;
37
38     if (priority == 1)
39     {
40         fine = fine * 0.80;
41     }
42 }
43
44 else if (bookType == 3)
45 {
46     fine = days * 30;
47 }
48
49 else
50 {
51     printf("Invalid Book Type");
52     return 0;
53 }
54
55 printf("Fine = Rs %.2f\n", fine);
56
57 if (bookType == 3 && days > 10)
58 {
59     printf("Banned from Borrowing");
60 }
61
62 return 0;
63 }
```

C:\Users\human\Documents\l x + v

```
Enter overdue days: 7
Enter book type (1=Regular, 2=Reference, 3=Rare): 1
Enter priority membership (1=Yes, 0=No): 0
Fine = Rs 35.00

-----
Process exited after 18.72 seconds with return value 0
Press any key to continue . . . |
```

Task 7 — The Cricket Team Selector

Write a C program that takes batting average as a float, matches played as an integer, and fitness failure status as an integer where 1 means failed and 0 means passed. A player is normally selected when the batting average is at least 35 and matches played are at least 10. A player with an average between 25 and 34.99 and at least 20 matches can still be selected through the "Experience Quota", but this exception does not apply if the player failed the fitness test; in that case the player must be "Rejected — Fitness". Any player with fewer than 5 matches must always be "Rejected — Insufficient Matches" regardless of other conditions, and this check must be performed first. Display exactly one of "Selected", "Selected (Experience Quota)", "Not Selected", "Rejected — Fitness", or "Rejected — Insufficient Matches".

```
Untitled1.cpp
3 int main()
4 {
5     float average;
6     int matches, fitness;
7
8     printf("Enter batting average: ");
9     scanf("%f", &average);
10
11     printf("Enter matches played: ");
12     scanf("%d", &matches);
13
14     printf("Enter fitness status (1=Failed, 0=Passed): ");
15     scanf("%d", &fitness);
16
17     if (matches < 5)
18     {
19         printf("Rejected - Insufficient Matches");
20     }
21     else if (fitness == 1)
22     {
23         printf("Rejected - Fitness");
24     }
25     else if (average >= 35 && matches >= 10)
26     {
27         printf("Selected");
28     }
29     else if (average >= 25 && average < 35 && matches >= 20)
30     {
31         printf("Selected (Experience Quota)");
32     }
33     else
34     {
35         printf("Not Selected");
36     }
37
38     return 0;
39 }
```

```
Enter batting average: 40
Enter matches played: 10
Enter fitness status (1=Failed, 0=Passed): 1
Rejected - Fitness
-----
Process exited after 12.71 seconds with return value 0
Press any key to continue . . .
```

Task 8 — The Restaurant Billing System

Write a C program that takes the meal category, bill amount, and customer type as inputs. The meal category is 1 = Fast Food, 2 = Desi Food, and 3 = Chinese, while customer type is 1 = Student and 2 = Regular. Use a switch statement to determine the service charge: Fast Food has 5%, Desi Food has 8%, and Chinese has 10%. After calculating the service charge, use nested ifelse to apply a discount: if the bill is Rs 1000 or more, students receive 15% and regular customers receive 10%; if the bill is below Rs 1000, students receive 5% while regular customers receive no discount. Display the service charge, discount, and final payable amount up to 2 decimal places. If the meal category or customer type is invalid, display "Invalid Selection"

```

1 #include <stdio.h>
2
3 int main()
4 {
5     int meal, customer;
6     float bill, serviceCharge, discount, finalBill;
7
8     printf("Enter bill amount: ");
9     scanf("%f", &bill);
10
11     printf("\nSelect Meal Type:\n");
12     printf("1. Fast Food\n");
13     printf("2. Desi Food\n");
14     printf("3. Chinese Food\n");
15     printf("Enter choice: ");
16     scanf("%d", &meal);
17
18     printf("\nSelect Customer Type:\n");
19     printf("1. Student\n");
20     printf("2. Regular Customer\n");
21     printf("Enter choice: ");
22     scanf("%d", &customer);
23
24     /* Service Charge using switch */
25     switch(meal)
26     {
27         case 1:
28             serviceCharge = bill * 0.05;
29             break;
30
31         case 2:
32             serviceCharge = bill * 0.08;
33             break;
34
35         case 3:
36             serviceCharge = bill * 0.10;
37             break;
38     }

```

```

39
40     /* Discount */
41     if(bill >= 1000)
42     {
43         if(customer == 1)
44             discount = bill * 0.15;
45         else if(customer == 2)
46             discount = bill * 0.10;
47         else
48         {
49             printf("Invalid customer choice!");
50             return 0;
51         }
52     }
53     else
54     {
55         if(customer == 1)
56             discount = bill * 0.05;
57         else if(customer == 2)
58             discount = 0;
59         else
60         {
61             printf("Invalid customer choice!");
62             return 0;
63         }
64     }
65
66     finalBill = bill + serviceCharge - discount;

```

```

67
68
69
70
71
72     printf("\nOriginal Bill = %.2f", bill);
73     printf("\nService Charge = %.2f", serviceCharge);
74     printf("\nDiscount = %.2f", discount);
75     printf("\nFinal Bill = %.2f", finalBill);
76
77     return 0;
78 }

```

```

C:\Users\humal\Documents\U...
Enter bill amount: 5000

Select Meal Type:
1. Fast Food
2. Desi Food
3. Chinese Food
Enter choice: 3

Select Customer Type:
1. Student
2. Regular Customer
Enter choice: 1

Original Bill = 5000.00
Service Charge = 500.00
Discount = 750.00
Final Bill = 4750.00
-----
Process exited after 21.18 seconds with return value 0
Press any key to continue . . .

```